PREMIER UNIVERSITY

Department of Computer Science & Engineering

CSE 4345 | COMPLEX ENGINEERING ASSIGNMENT

EpiScope

Big Data Surveillance, Forecasting and Response-Planning Platform for Seasonal, Tropical and EpidemicDiseases

GROUP MEMBERS AND STUDENT IDs

Name	Student ID
Prionti Dey	0222220005101009
Satyajit Mohajan	0222220005101010
Pranta Chowdhury	0222220005101011
Borshan Karmaker	0222220005101020

GITHUB REPOSITORY URL

[**https://github.com/satyajit-mohajan/EpiScope**]

SHAREABLE VIDEO PRESENTATION URL

[**https://drive.google.com/drive/folders/1e_r2lhz-eWV1iMAHl0QXKLOmFWV9DLzM**]

01 / PROJECT OVERVIEW

Executive summary

A different approach to the same EpiScope brief: emphasize explainable early signals, week-level forecasting, and scenario literacy.

Research aim. EpiScope Sentinel is a standalone, local-first teaching platform that investigates whether syndromic activity may look atypical, estimates near-term aggregate counts, and makes intervention assumptions explicit. The project does not diagnose patients, detect confirmed outbreaks, or prescribe a response.

Engineering objectives

- Produce a reproducible, explicitly synthetic weekly multi-location surveillance dataset.
- Validate records and calculate readable four-week trends and abnormality flags.
- Evaluate weekly Poisson count forecasts against a time-respecting naive baseline.
- Allow users to contrast a model baseline with a non-causal reduction scenario.
- Publish a browser dashboard, documented JSON API and repeatable automated tests.

How Sentinel differs from the earlier EpiScope

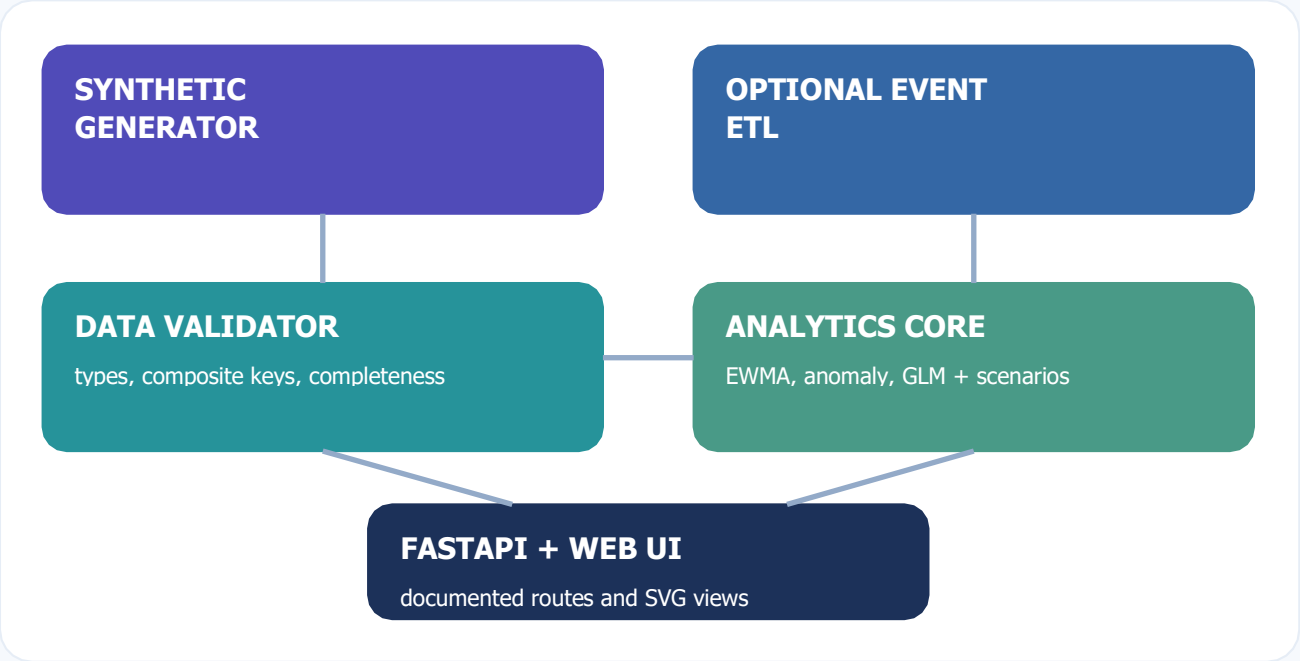
- **Primary unit:** epidemiological weeks, rather than day-by-day surveillance.
- **Analytical focus:** four-week anomalies and EWMA patterns, instead of only risk ranks.
- **Forecast model:** regularized Poisson GLM with Fourier seasonality, not autoregressive Ridge.
- **Planning:** hypothetical effect simulator rather than optimization of a constrained budget.
- **Interface:** top-navigation visual lab with lightweight SVG graphs and fresh architecture.

Scope statement: A functioning educational prototype and reproducible benchmark. The included 4,752 records are **not Big Data by volume**; separate streaming and optional Spark samples illustrate scalable patterns.

02 / ARCHITECTURE

System architecture

Clear module boundaries allow validation, analytics, forecasting, scenario exploration and presentation to evolve separately.



Design decisions

- **Local-first:** FastAPI serves all frontend assets without external chart libraries or cloud accounts.
- **Observable calculations:** each endpoint returns the synthetic label and interpretable method outputs.
- **Reproducibility:** fixed random seed; quality rules applied before any analytics.
- **Extension point:** independent event-streaming exercise demonstrates data engineering beyond the core CSV.

03 / DATA ENGINEERING

Synthetic data & validation

A complete demonstration schema with explicit provenance, consistent keys and zero included patient identifiers.

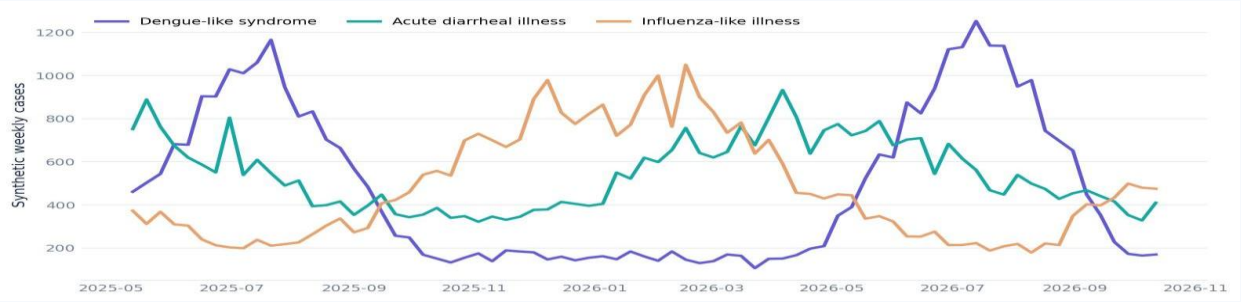
RECORDS	DISTRICTS	SYNDROMES	WEEKS
4752	8	3	198

Weekly observation schema

week_start	Monday ISO date
district	8 named demonstration areas
syndrome	dengue / diarrhea / ili
cases	nonnegative integer
visits	visits >= cases
source	SIMULATED_SENTINEL

Simulation patterns

Three independent syndrome-like signals have deliberately different annual shapes; observations are generated by negative-binomial draws with local multipliers and a small serial contribution. Three brief spikes are injected at predeclared simulated locations and weeks.



Important: No included cases, population figures or syndrome-level visit ratios represent real Bangladesh disease trends. The optional encounter generator creates a separate engineering demo, not extra surveillance evidence.

04 / WARNING SIGNALS

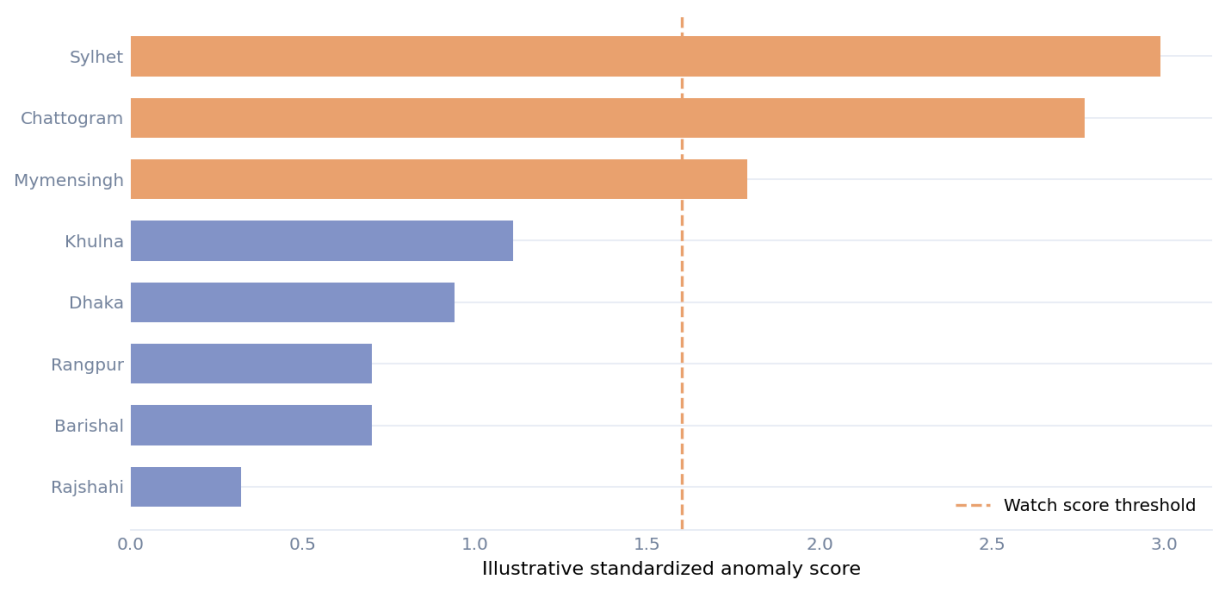
Explainable detection engine

A deliberately interpretable rule combines an anomaly statistic with four-week directional change.

Reference: 26 weeks preceding the recent four weeks. **Current:** mean of the last four weeks. Standardized score uses the older-window standard deviation with a floor to stabilize low counts. EWMA with alpha = 0.35 is displayed as a secondary smoothing statistic.

WATCH: z-score >= 1.6 AND four-week momentum >= +5%.
HIGH: z-score >= 3.0 AND four-week momentum >= +20%.
NORMAL: all other combinations.

Example output: influenza-like illness



Observed simulated watch flags: Sylhet, Chattogram, Mymensingh. Flag statuses depend on the synthetic data cutoff of 2026-10-12 and may change if the data generator is rerun with a different seed.

Caveat: The rule thresholds are classroom choices and lack sensitivity, specificity or real-world outbreak validation. A signal is a visual prompt for exploration, not an instruction to issue public warnings.

05 / COUNT FORECASTING

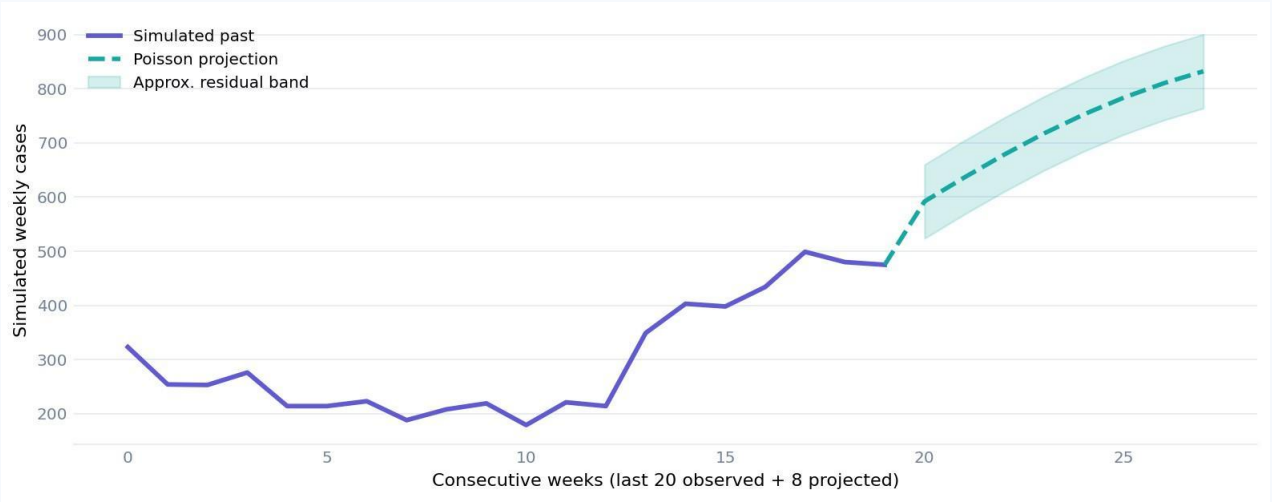
Poisson model & evaluation

Seasonal Fourier features and a chronological holdout make the comparison visible and resistant to simple future leakage.

Model: regularized Poisson regression (log-link) with annual sin/cos terms at harmonics 1 and 2 plus bounded time trend. **Evaluation:** train on all weeks except the most recent 12; predict those 12 and compute mean absolute error (MAE). **Comparator:** count from exactly 52 weeks earlier.

Syndrome	Poisson MAE	Seasonal naive MAE
Dengue-like	**68.45**	**67.92**
Diarrheal illness	**42.92**	**46.50**
Influenza-like	**40.63**	**58.17**

Eight-week outlook (ILI, all districts)



Transparent result: For dengue, Poisson MAE 68.45 is marginally **worse** than the seasonal-naive 67.92. The model is not presented as universally superior.

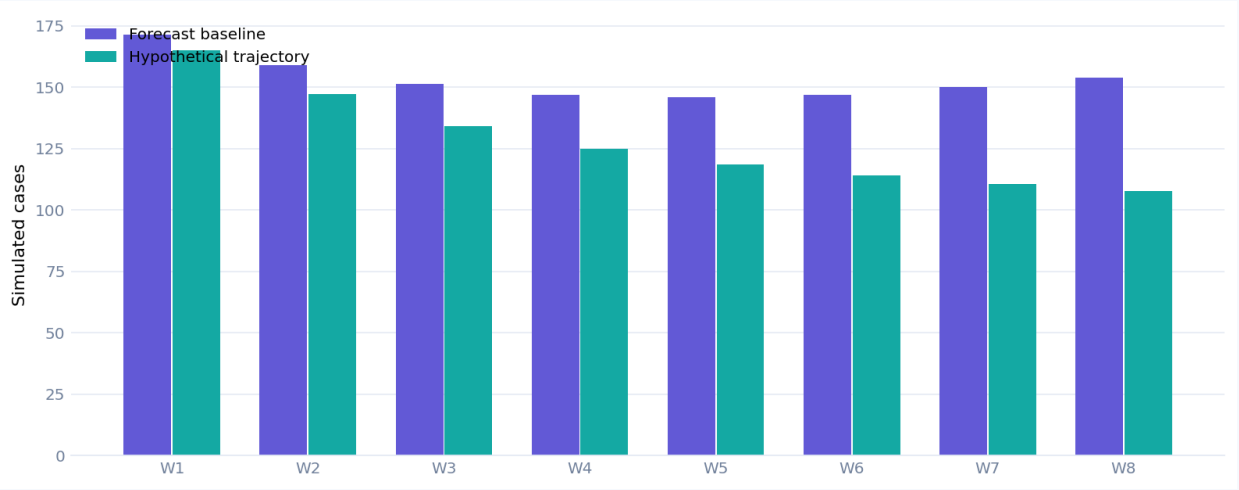
Uncertainty note: the displayed +/- 80th-percentile absolute holdout residual band is descriptive, not a statistically calibrated 80% prediction interval. The empirical holdout coverage for each syndrome was 75% in this small sample.

06 / RESPONSE PLANNING

Hypothetical intervention studio

A what-if calculator promotes transparent assumptions rather than false claims of treatment efficacy.

Let **r** be the selected final relative reduction, **H** the number of future weeks, and **t** the week offset. The simulator assumes **hypothetical_cases(t) = baseline_forecast(t) * [1 - r * t / H]**. The visualized difference is the arithmetic gap between two assumptions.



8-WEEK FORECAST BASELINE
1226

30% FINAL-WEEK HYPOTHESIS
1023

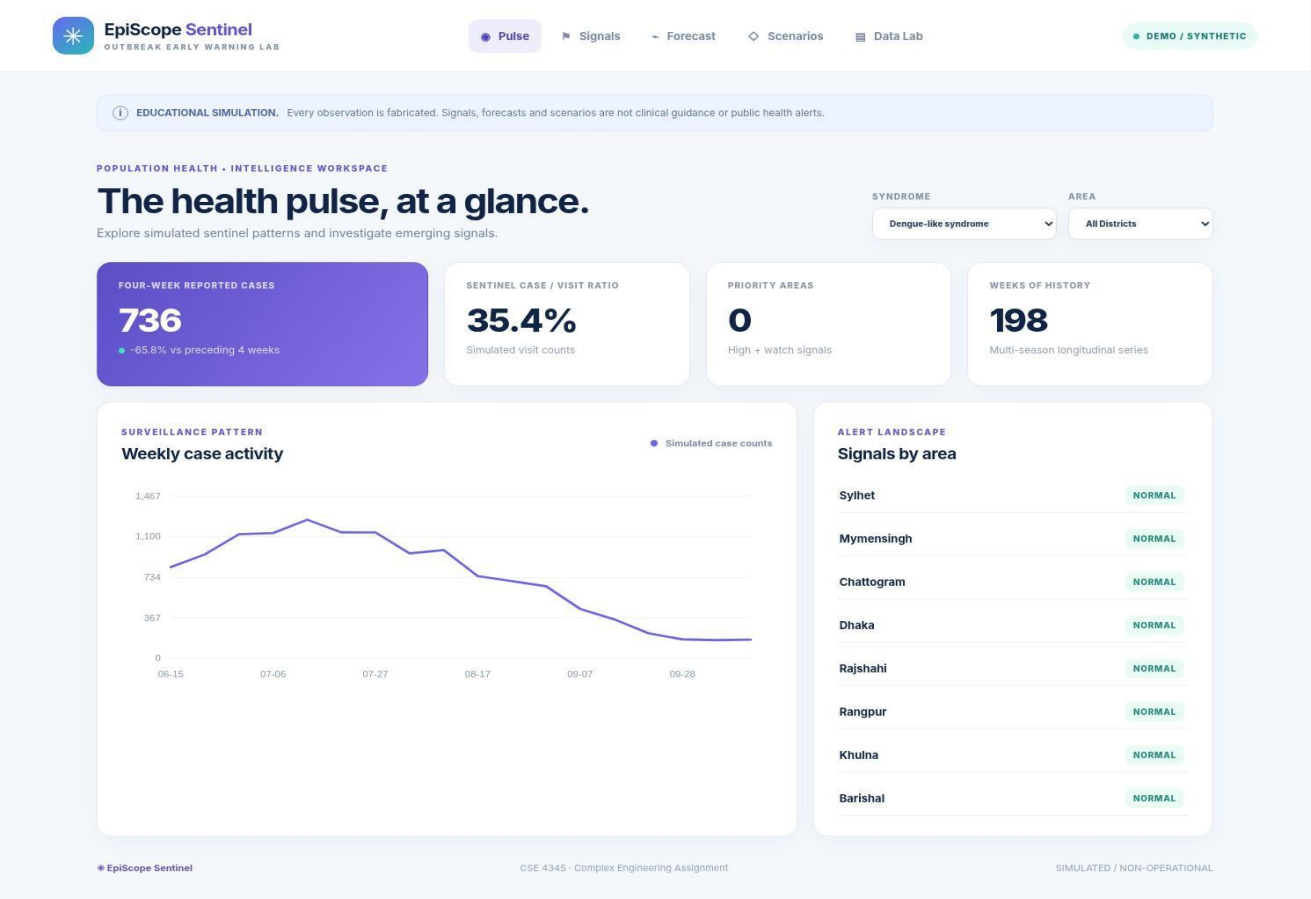
Interpretation & constraints

- The selected 30% number is hypothetical; it is **not** supported by efficacy trials or intervention data.
- The engine does not simulate transmission, social behavior, mobility or healthcare capacity.
- The same arithmetic structure can support discussion of awareness, sanitation or vector-control strategies without claiming effectiveness.
- Counterfactual totals are planning illustrations only: do not use for medical or government allocation decisions.

07 / PROTOTYPE INTERFACE

Product implementation

Accessible top navigation separates live indicators, anomaly exploration, forecast validation and scenario reasoning.



Pulse dashboard: four-week cases, simulated case-to-visit ratio, count of flagged areas and history length. It uses lightweight SVG graphics and does not need an external dashboard framework, an internet connection or patient information.

API surfaces

/api/overview	case burden, visit ratios and flags
/api/signals	anomaly and EWMA per area
/api/forecast	8-week Poisson outlook and holdout MAE
/api/scenario	arithmetic what-if trajectories
/api/quality	provenance and dataset checks
/api/export	synthetic CSV download

08 / TESTS & ETHICS

Verification and constraints

A working teaching prototype must say what has been tested - and which safety properties are still missing.

AUTOMATED TESTS	FAILED TESTS
12	0

Automated verification covers

- Dataset completeness (4,752 rows), dates, unique keys and source labeling.
- Validation rejects duplicate series entries and false provenance labels.
- Simulated spike produces appropriate high alert under the declared rule.
- Poisson forecasts are nonnegative with calculable chronological holdout MAE.
- Scenario outcomes are monotonic with the selected hypothetical reduction.
- Streaming fake-event aggregation conserves a 1,017-event test input.
- Web assets, JSON API outputs, CSV export and invalid query handling.

Ethics, privacy and production gaps

Privacy: no identifiers or real patient files are stored. **Governance:** no real data use or health-ethics clearance is implied. **Security:** local prototype has no authentication, patient consent flow, encryption policy, rate limits or audit logging. **Accuracy:** flags, forecast intervals and scenario effects require extensive field evaluation before public-health adoption.

Scaling caveat: the main CSV is small. Streaming event samples and optional Spark ETL show architectural techniques only; distributed production throughput was not measured.

09 / CONCLUSION

Findings, handover & next steps

A self-contained research prototype that can be run, explained, tested and extended.

Summary of contributions

- A fully simulated weekly sentinel dataset and validation pipeline.
- Transparent anomaly indicators and a directly inspectable threshold policy.
- Seasonality-aware Poisson count forecast with an honest naive comparison.
- An interactive, explicitly non-causal hypothetical scenario lab.
- A documented local API, responsive web interface, and twelve passing tests.

How to run and submit

- Extract ZIP; on Windows double-click **START_WINDOWS.bat** (Python 3.10+ required).
- Open **http://127.0.0.1:8000** and **/docs** for the API.
- Install dev dependencies and run **python -m pytest -q** to verify locally.
- Upload source folders to the friend group's own GitHub repository.
- Replace **all cover placeholders**, then submit ZIP/repo URL, PDF and shareable recorded presentation link.

Directions for future research

Secure authorized epidemiological feeds; external weather and mobility covariates; calibrated count intervals; geographic equity analysis; cross-validated threshold selection; encrypted role-based data access; and production-scale stream processing.

Technical reading

- scikit-learn: PoissonRegressor API documentation and generalized linear models.
- FastAPI: Request validation, interactive API documentation and TestClient.
- Apache Spark: SQL and DataFrame documentation (optional ETL demonstration).
- Project README, DATA_DICTIONARY.md, METHODS.md and TEST_RESULTS.md.